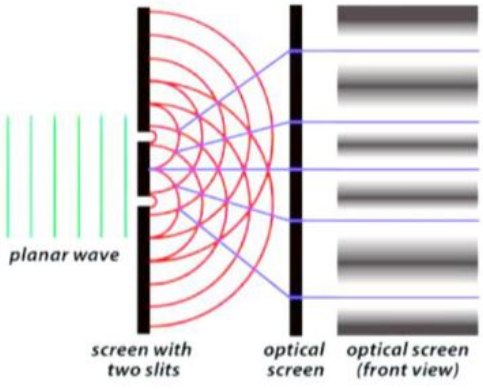
4(a)(ii)	example of transverse waves: light, EM waves, any EM waves example of longitudinal waves: sound	[1]
4(a)(i)	transverse waves: oscillations/vibrations of wave particles are perpendicular to the direction of energy transfer/transmission/travel longitudinal waves: oscillations/vibrations of wave particles are parallel to the of energy transfer/transmission/travel.	[1] [1]
4(b)(i)	Diffraction refers to the spreading of a wave when it passes through a slit (opening/gap/aperture), or around an obstacle (or around an edge) into the "shadow" region.	[1]
4(b)(ii)	 diffraction/spreading through each slit/slits allows overlap/meet/superpose of secondary wavelets for superposition/interference to occur.	[1] [1]

4(c)(i)	DC =**18 cm**, DABC =**80 cm**	[1]
4(c)(ii)	path difference = 80 -18 = 62 cm = 15½ wavelengths { as λ = 4.0 cm} phase difference =π radians or 180^0 resultant amplitude is a minimum. { Accept: zero }	[1] [1] [1, ecf]